Development of Sea Turtle Re-Identification Model using Vision Transformers

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Abstract*—Photo-based individual identification is an attractive non-invasive alternative to satellite tagging for sea turtle conservation, but published systems are typically reported as a single tuned configuration, leaving practitioners without guidance on which design choices actually matter. We fine-tune a DINOv3-pretrained Vision Transformer with an embedding head on SeaTurtleIDHeads under a strict open-set identity split, combining ArcFace and online hard triplet losses with layer-wise learning rate decay. We then vary nineteen parameters one factor at a time to build a sensitivity map. A conventionally chosen baseline reaches Rank-1 of 78.3% and mAP of 84.9%; combining every individually-best setting reaches 90.4% and 93.5%. Three findings are of practical interest. First, the backbone learning rate dominates: raising it alone from 10^{-5} to 10^{-4} yields Rank-1 of 87.8%, roughly 78% of the total gain, whereas six of the nineteen parameters move Rank-1 by under two points. Second, parameter effects are strongly sub-additive: summing the individual gains predicts $+33.7$ points, an arithmetically impossible figure, while only $+12.1$ points materialize. Third, and most consequential for deployment, the embedding transfers to the unseen SeaTurtleID2022 domain at Rank-1 of 81.1%, yet open-set rejection of unenrolled individuals collapses the effective accuracy to 55–57% at a 42% false-reject rate. Two structurally different rejection rules converge on nearly identical operating points because all top-1 similarities occupy a narrow 0.85–0.98 band with ROC-AUC of only 0.77–0.79. The bottleneck is score calibration, not representation quality.

***Index Terms*—animal re-identification, vision transformer, DINOv3, metric learning, open-set recognition, wildlife conservation, sea turtle**

I. Introduction

Sea turtles are globally threatened, and effective conservation depends on tracking individuals across time to estimate population size, migration routes, and reproductive behaviour. The dominant instrumented approach, satellite tagging, is expensive per unit, prone to device failure, and invasive enough to require careful attachment to avoid harming the animal [1].

Photo-identification offers a non-invasive alternative. The keratin scutes covering a turtle's head form a geometric pattern that develops embryonically and does not structurally regenerate, making it a stable natural biometric [2]. Automated re-identification (Re-ID) over such patterns has progressed from hand-crafted local descriptors [3], [4] to convolutional networks [5] and, most recently, to large self-supervised backbones [6].

Re-ID is not closed-set classification. The model must map images to an embedding space in which the same individual is close and different individuals are far apart, and it must do so for identities never seen during training [7]. This open-set requirement is precisely what makes the task useful in the field, where new animals are encountered continuously.

Existing sea turtle Re-ID work reports a single tuned system. That is sufficient to establish feasibility, but it does not tell a practitioner adapting the pipeline to a new site which knobs to turn first, nor how much of the reported accuracy survives a change of camera and location. This paper addresses both gaps. Our contributions are:

- A DINOv3 ViT Re-ID pipeline for loggerhead head images trained under an open-set identity split, evaluated with and without k-reciprocal re-ranking.
- A sensitivity map over nineteen parameters obtained by one-factor-at-a-time (OFAT) variation from a deliberately convention-based, not result-tuned, baseline. This ordering is, to our knowledge, the first published for animal Re-ID with ViTs.
- An explicit demonstration that OFAT gains do not compose. Summing individual improvements over-predicts the combined configuration by a factor of roughly 2.8, and the predicted value exceeds 100% Rank-1.
- A cross-dataset and open-set evaluation on SeaTurtleID2022 showing that representation quality transfers but rejection of unenrolled individuals does not, and localizing the failure to the similarity score distribution rather than to the embedding.

We deliberately report the negative open-set result in full. A conservation system that silently rejects two of every five valid

sightings is not deployable, and identifying why is more useful than reporting only the favourable in-distribution number.

II. Related Work

A. Photo-Identification of Sea Turtles

Carter et al. [8] applied neural network ensembles to green turtle facial scutes, marking the shift from expert-dependent manual matching toward reproducible automation. Carpentier et al. [2] validated facial scute stability over an eleven-year span, supplying the biological justification for head-based identity systems. Gatto et al. [3] and Dunbar et al. [4] explored front-flipper scutes as an alternative region, with the latter reporting higher accuracy for flippers under certain conditions, though head images remain preferred in the field for ease of capture.

B. Deep Learning Benchmarks

Adam et al. [9] introduced SeaTurtleID2022, comprising 8,729 photographs of 438 loggerhead individuals collected over thirteen years in the Greek islands, along with a time-aware open-set protocol and a Hybrid Task Cascade plus ArcFace baseline reaching Rank-1 of 86.8%. Nguyen et al. [5] benchmarked CNN architectures on facial scutes, reaching 95.3% with ResNet-50 on a mixed-quality set of 1,426 images across 20 identities. Čermák et al. [6] released the WildlifeDatasets toolkit and MegaDescriptor, a multi-species Re-ID foundation model that outperforms generic pretrained backbones including CLIP and DINOv2. Papafitsoros et al. [10] exploited visual similarity between left and right facial profiles of the same individual, a signal conventional photo-ID pipelines discard by partitioning libraries per side.

We emphasize that our numbers are not directly comparable to [9] or [5]: the datasets, identity counts, and split protocols differ. Our contribution is the relative structure of the design space, which is comparable across setups in a way that absolute accuracy is not.

C. Self-Supervised Vision Transformers

The Vision Transformer [11] processes an image as a sequence of patches, allowing every patch to attend to every other at each layer. This suits Re-ID of scute patterns, whose discriminative regions are spatially scattered across the head. DINO [12] showed that self-distillation without labels induces attention maps containing object boundary information at pixel resolution; DINOv2 [13] scaled this to a curated 142M-image corpus with combined image- and patch-level objectives; DINOv3 [14] further improves dense feature correspondence and out-of-distribution generalization. Boundary-sensitive dense features are directly relevant here, since no scute-level segmentation annotation exists for turtles.

D. Metric Learning Objectives

ArcFace [15] applies an additive angular margin on the unit hypersphere, giving a margin with consistent geodesic meaning regardless of class position. Triplet loss [16] with batch-hard mining [17] optimizes relative distances directly. Their linear combination is standard in competitive Re-ID, on the premise that ArcFace provides global inter-class separation while triplet loss enforces local distance consistency. One of our findings challenges that premise at this data scale.

III. Methodology

A. Dataset and Open-Set Protocol

We train on SeaTurtleIDHeads [22], containing 7,582 head-cropped images of 400 loggerhead (*Caretta caretta*) individuals photographed over more than a decade in Greek waters. The set exhibits wide variation in pose, illumination, viewing angle, camera distance, and resolution, reflecting realistic field capture. We prefer it over SeaTurtleID2022 as the training source because the latter has lower image resolution and was designed primarily for segmentation.

The dataset is partitioned 70:20:10 into train, validation, and test by *identity*, so no individual appears in more than one partition. All reported evaluation is therefore open-set at the identity level. Identities with fewer than $N_{\min} = N_G + 2$ images are discarded so that every retained identity can supply a gallery and at least two queries. Validation and test partitions are each split into a gallery of N_G randomly chosen images per identity and a query set of the remainder. Validation is used only for early stopping and checkpoint selection; test is touched once.

B. Preprocessing and Augmentation

Images are resized to `img_size` $\times$ `img_size`. Training augmentation comprises random crop from a $+32$ pixel resize, horizontal flip ($p = 0.5$), colour jitter over brightness, contrast, saturation and hue, random grayscale ($p = 0.1$), rotation up to $20°$, Gaussian blur with a 5×5 kernel and $\sigma \in [0.1, 2.0]$, and random erasing ($p = 0.3$). Evaluation applies resize and normalization only, keeping results deterministic. Pixel statistics follow ImageNet conventions, matching the pretraining regime.

C. Architecture

The backbone is a DINOv3-pretrained ViT loaded from `timm`, with a 16×16 patch grid. At 256×256 input this yields 256 patch tokens plus one class token. The class token output of dimension 768 (ViT-B) is projected by an embedding head consisting of layer normalization, dropout with probability p_{drop}, and a bias-free linear map to dimension d_e. The output is L_2-normalized, placing all embeddings on the unit hypersphere so that cosine similarity reduces to a dot product.

D. Objective

ArcFace loss is the primary term:

$$\mathcal{L}_{\text{Arc}} = -\frac{1}{N}\sum_{i=1}^{N} \log \frac{e^{s_a \cos(\theta_{y_i} + m_a)}}{e^{s_a \cos(\theta_{y_i} + m_a)} + \sum_{j \neq y_i} e^{s_a \cos \theta_j}} \quad (1)$$

where θ_{y_i} is the angle between the sample embedding and its class prototype. An online hard triplet term selects, for each anchor in the batch, the farthest positive and nearest negative:

$$\mathcal{L}_{\text{Tri}} = \frac{1}{N}\sum_{i=1}^{N} \max\big(d(a_i, p_i) - d(a_i, n_i) + m_t,\ 0\big) \tag{2}$$

with masked selection to prevent zero-valued non-pair entries from being mistaken for the smallest positive distance. The total objective is

$$\mathcal{L} = \mathcal{L}_{\text{Arc}} + \lambda_t \mathcal{L}_{\text{Tri}}. \tag{3}$$

E. Optimization

Batches are formed by a balanced sampler drawing P identities and K images per identity, guaranteeing valid positive pairs. The optimizer is AdamW with decoupled weight decay λ_w. Layer-wise learning rate decay assigns transformer block i of L a rate

$$\eta_i = \eta_{\text{bb}} \cdot \gamma^{\,L-i}, \tag{4}$$

so shallower blocks, which encode more generic features, are updated more conservatively. The randomly initialized embedding head and ArcFace layer use a separate, higher base rate η_{base}. A LambdaLR schedule applies linear warmup over W epochs followed by cosine decay to zero over the remaining $E - W$ epochs. Gradients are clipped at $G_{\max}$ and mixed precision is enabled.

F. Inference and Evaluation

Embeddings are extracted in a single forward pass over the resized image and L_2-normalized; no test-time augmentation is used, so the same protocol applies to in-distribution and cross-dataset evaluation. Gallery images of one identity are mean-pooled into a single prototype and renormalized. Similarity is computed as $S = QG^{\top}$. We optionally apply k-reciprocal re-ranking [18], which combines cosine and Jaccard distances,

$$D_{\text{final}}(q, g) = \lambda_{rr} D_{\cos}(q, g) + (1 - \lambda_{rr}) D_{\text{Jac}}(q, g), \tag{5}$$

with $k_1 = 20$, $k_2 = 6$, $\lambda_{rr} = 0.3$, accelerated by FAISS. We report CMC at ranks 1, 5, 10 and mAP.

IV. Experiments

A. Experimental Setup

Experiments run on an NVIDIA RTX 4090, Intel Core i9-13900KF, 64 GB RAM, Python 3.10, PyTorch 2.11 with CUDA 12.6, and `timm` 1.0.25. A fixed seed of 42 is set across `random`, `numpy`, and `torch`.

The baseline in Table I was fixed *before* any variation experiment, from conventions in the ViT and metric-learning literature adjusted for dataset scale. This matters methodologically: had the reference point been chosen using the variation results, the OFAT analysis would be circular and no parameter could be shown to be mis-set. The conservative backbone learning rate that convention prescribes turns out to be the single largest performance limiter, a finding only reachable because the baseline was not tuned.

TABLE I
BASELINE CONFIGURATION

Group	Parameter	Value
Data	Image size	256
	Gallery images per identity N_G	3
Architecture	Backbone	ViT-B/16 DINOv3
	Embedding dim d_e	256
	Dropout p_{drop}	0.1
Loss	ArcFace scale s_a	32.0
	ArcFace margin m_a	0.5
	Triplet margin m_t	0.3
	Triplet weight λ_t	0.5
Training	Identities per batch P	16
	Images per identity K	4
	Epochs E	50
	Warmup epochs W	5
	Gradient clip $G_{\max}$	1.0
	Early-stop patience P_{stop}	10 intervals
Optimizer	Weight decay λ_w	10^{-4}
	Base LR η_{base}	10^{-3}
	Backbone base LR η_{bb}	10^{-5}
	LLRD factor γ	0.75

TABLE II
BASELINE EVALUATION ON THE SEATURTLEIDHEADS TEST SPLIT

Metric	Cosine	+Re-rank	Δ (pp)
Rank-1	77.4%	78.3%	+0.9
Rank-5	91.5%	92.3%	+0.8
Rank-10	95.9%	95.1%	−0.8
mAP	83.6%	84.9%	+1.3

V. Results

A. Baseline Performance

Training loss decreases monotonically across all 50 epochs without spikes, indicating that five warmup epochs and clipping at 1.0 suffice for stability. Validation metrics rise sharply during warmup and plateau under cosine decay with no late-epoch decline. The absence of overfitting at $E = 50$ is consistent with two later findings: extending to $E = 100$ still improves performance, and adding regularization degrades it. The model is budget-limited, not regularization-limited.

Table II shows a steep Rank-1 to Rank-5 gain of 14.0 points against only 2.8 from Rank-5 to Rank-10. For roughly 14% of queries the correct identity is already within the top five but loses narrowly at rank one, so most failures are ordering errors rather than retrieval failures. Qualitative inspection shows the residual 4.9% concentrated on blurred, low-contrast images where scute geometry is no longer legible.

Re-ranking helps at the top of the list and on mAP but slightly hurts Rank-10, which is expected: it optimizes neighbourhood consistency and occasionally pushes a correct candidate out of a deeper cutoff. Net effect is positive but modest, on the order of one point.

TABLE III
PARAMETER SENSITIVITY, ORDERED BY RANK-1 RANGE. BASELINE RANK-1 = 78.3%, ALL VALUES WITH RE-RANKING

Param.	Base	Best	Best R-1	Range	
η_{base}	10^{-3}	10^{-3}	78.3%	25.2	
N_G	3	3	78.3%	20.1[†]	
η_{bb}	10^{-5}	10^{-4}	**87.8%**	19.6	
backbone	ViT-B/16	ViT-L/16	80.4%	19.0	
E	50	100	84.4%	14.7	
P	16	8	79.1%	10.6	
p_{drop}	0.1	0.0	81.0%	7.9	
K	4	2	79.1%	7.2	
λ_t	0.5	0.0	83.4%	5.8	[†] Varying N_G changes
s_a	32.0	32.0	78.3%	5.4	
img. size	256	256	78.3%	5.0	
m_a	0.5	1.0	80.0%	3.5	
W	5	3	81.0%	2.7	
d_e	256	256	78.3%	1.8	
m_t	0.3	0.3	78.3%	1.8	
$G_{\max}$	1.0	1.0	78.3%	1.7	
γ	0.75	0.65	79.5%	1.2	
P_{stop}	10	20	78.5%	0.9[‡]	
λ_w	10^{-4}	10^{-5}	79.1%	0.8	

the query/gallery composition, so rows are not evaluated on an identical query set.
[‡] Early stopping cannot trigger at $P_{\text{stop}} \geq 10$ under $E = 50$; the three largest values effectively disable it.
All values from a single seed; between-seed variance was not measured.

B. Parameter Sensitivity

Table III ranks all nineteen parameters by Rank-1 range, the spread between the best and worst tested value. Range measures how far performance moves when a parameter is misconfigured, independent of whether the baseline was already correct; the Δ column separately reports the headroom available above the baseline.

Three groups emerge.

Sensitive but already correct. η_{base}, s_a, and image size degrade in either direction. The extreme case is η_{base}: a one-decade shift either way costs more than 22 points, since the randomly initialized head fails to converge when the rate is too high and fails to learn within budget when too low. These confirm the convention-based choices were right, and that getting them wrong is expensive.

Real headroom. η_{bb}, E, λ_t, p_{drop}, backbone, m_a, and W. Raising η_{bb} to 10^{-4} alone gives Rank-1 of 87.8% and mAP of 91.6%. Because the trend is monotonic and 10^{-4} is the upper bound tested, the true optimum may lie higher. Notably $\lambda_t = 0$ is best, i.e. disabling triplet loss entirely gives Rank-1 of 83.4% with performance decreasing monotonically in λ_t. At this data scale ArcFace alone is sufficiently discriminative and the triplet term interferes with optimization. This also removes the justification for PK sampling, which explains why $K = 2$ outperforms $K = 4$.

Batch composition is better read through effective batch size than through P or K separately: both size-32 settings give identical Rank-1 of 79.1% and both size-128 settings collapse to 68.5–71.9%. Under a fixed epoch budget, larger batches quarter the number of update steps, so this is an under-training symptom, not a diversity effect.

Negligible. d_e, m_t, $G_{\max}$, γ, P_{stop}, λ_w all move under two points, which on a query set of a few hundred amounts to a handful of queries. With a single seed these cannot be distinguished from noise, so we claim insensitivity rather than optimality for this group. The P_{stop} result is structural rather than empirical: at $E = 50$ with evaluation every five epochs, $P_{\text{stop}} \geq 10$ never fires.

TABLE IV
BASELINE VERSUS COMBINED CONFIGURATION

	Cosine		+Re-rank	
Config	R-1	mAP	R-1	mAP
Baseline	77.4%	83.6%	78.3%	84.9%
$\eta_{\text{bb}} = 10^{-4}$ only	—	—	87.8%	91.6%
Combined	89.1%	92.0%	**90.4%**	**93.5%**

C. Combined Configuration and Sub-Additivity

The combined configuration adopts every individually-best value from Table III with positive Δ: ViT-L/16, $p_{\text{drop}} = 0$, $m_a = 1.0$, $\lambda_t = 0$, $P = 8$, $K = 2$, $E = 100$, $W = 3$, $P_{\text{stop}} = 20$, $\lambda_w = 10^{-5}$, $\eta_{\text{bb}} = 10^{-4}$, $\gamma = 0.65$; twelve changes in total.

The combined configuration reaches Rank-1 of 90.4% and mAP of 93.5%, gains of 12.1 and 8.6 points. The interesting comparison is against the additive prediction. Summing the Δ column gives $+33.7$ points, predicting $78.3 + 33.7 = 112.0\%$ Rank-1. The analogous mAP calculation predicts 105.6%. Both exceed the theoretical ceiling, so additivity is not merely unsupported empirically but arithmetically incoherent: only 21.7 points of headroom exist while the sum demands 33.7.

Realized gains are 36% of predicted on Rank-1 and 42% on mAP. The consistency of this ratio across two metrics indicates a systematic property rather than a metric artifact. Two overlap groups explain it. First, η_{bb} ($+9.5$), E ($+6.1$), W ($+2.7$), ViT-L ($+2.1$), and γ ($+1.2$) all address the same limitation, namely insufficient backbone adaptation to the turtle domain; once η_{bb} is raised, the deficit that $E = 100$ would fix is largely already resolved. Second, λ_t ($+5.1$), p_{drop} ($+2.7$), and λ_w ($+0.8$) are all relaxations of regularization on a model that was never overfitting, so they harvest the same surplus. Several Δ values are below one point and plausibly noise.

Critically, $\eta_{\text{bb}} = 10^{-4}$ alone accounts for 9.5 of the 12.1 point total, about 78%; the other eleven changes contribute 2.6 points collectively. Backbone learning rate tuning is not merely the most influential parameter but very nearly the only one with substantial effect on this system.

The practical consequence is that concatenating OFAT optima does not guarantee an optimum and can import harmful changes. One is already visible: $\gamma = 0.65$ was selected on Rank-1 evidence ($+1.2$) despite reducing mAP by 0.2 and Rank-5 by 2.1 points in its own OFAT run. Verification by factorial design, or minimally by backward ablation from the combined configuration, is required.

D. Cross-Dataset Generalization

We apply the combined-configuration checkpoint to SeaTurtleID2022 [9] without retraining. Twenty-five identities are

TABLE V
CROSS-DATASET RETRIEVAL ON SEATURTLEID2022, NO RETRAINING, NO REJECTION

Quantity	Value
Registered identities	25
Gallery images per identity N_G	3
Queries	217
Rank-1	81.1%
Rank-5	97.2%

TABLE VI
OPEN-SET IDENTIFICATION ON SEATURTLEID2022 (217 REGISTERED, 49 UNREGISTERED QUERIES)

Quantity	Global thr.	Background
Decision parameter	0.9122	+0.0148
Identification accuracy	63.5%	64.3%
Registered Rank-1	55.3%	56.7%
False-reject rate	43.4%	41.9%
Unregistered recall	100.0%	98.0%
False-accept rate	0.0%	2.0%

drawn at random as registered, with $N_G = 3$ gallery images each and the remainder as queries; images are head-cropped to match training preprocessing. Since the inference protocol is identical to the in-distribution evaluation, any degradation is attributable to domain shift alone.

Rank-1 of 81.1% is 9.3 points below the in-distribution combined result, so transfer carries a measurable cost, but it is not catastrophic. Rank-5 of 97.2% in fact exceeds the in-distribution combined Rank-5 of 96.6%, and cross-dataset Rank-1 still exceeds the in-distribution *baseline*. The fine-tuned DINOv3 embedding does generalize across capture conditions. t-SNE projections show clusters that remain separable but are looser intra-class and closer inter-class than in-distribution, consistent with the observed drop.

E. Open-Set Rejection

Field deployment must handle photographs of animals never enrolled. Adding five unregistered identities brings the query set to 266, comprising 217 registered and 49 unregistered. We compare two rejection rules on the identical query set.

Global threshold (absolute). The top-1 cosine score is compared against a single threshold tuned by maximizing Youden's J.

Background class (relative). A set of background prototypes drawn from turtle images outside the gallery, kept as multiple prototypes rather than pooled, yields a relative score $s = s_{\text{reg}} - s_{\text{bg}}$, where s_{bg} is the maximum similarity to any background prototype. A query is accepted if $s \geq$ margin, with the margin tuned under a bounded false-reject constraint. Recomputing s_{bg} per query is intended to normalize out hub identities.

Registered Rank-1 falls from 81.1% without rejection to 55.3% and 56.7% respectively. The ranking model is unchanged between these settings; only a rejection stage placed before ranking has been added. The source is the false-reject rate of roughly 42%: nearly half of genuinely enrolled queries are discarded before they can be ranked.

The important observation is not which rule wins but how little separates them. Despite mechanistically opposite designs, absolute versus relative, they converge on nearly the same operating point, with all differences on the order of one or two queries. Among accepted registered queries, retrieval accuracy is $0.567/(1-0.419) \approx 97.6\%$ for the background scheme and $\approx 97.7\%$ for the threshold, confirming that both rules simply discard the hard cases first.

The cause lies in the score distribution. Every top-1 similarity, registered or not, falls in a narrow band of approximately 0.85 to 0.98. An image of a turtle absent from the gallery still scores about 0.90 against some other individual's prototype, because all images share nearly identical structure, a turtle head against an underwater background, placing embeddings in a tight cone on the hypersphere where any two images are already similar before identity is considered. The unregistered distribution is not merely shifted but embedded within the lower tail of the registered distribution, so any threshold high enough to reject all unregistered queries necessarily rejects most registered ones. The 43.4% false-reject rate is a direct consequence of this geometry, not of poor threshold tuning; the Youden optimum is a corner solution chosen because no better compromise exists.

For the background scheme, plotting s_{reg} against s_{bg} shows the two classes separating along the horizontal axis only. On the vertical axis they overlap almost completely, and s_{bg} shows no strong dependence on s_{reg}. The relative score is therefore effectively s_{reg} plus noise, which is why it reaches the same operating point. Overall separability is only moderate, with ROC-AUC of 0.77 to 0.79, and this bounds every possible decision rule over these scores.

Qualitative inspection is decisive. In the five most confidently wrong decisions, all five are registered identities rejected as unknown, none are false accepts, and in every case the correct identity remains within the top five candidates, at rank one for three of them with scores of 0.853, 0.867, and 0.868. Meanwhile a correctly rejected unregistered query scores 0.851. A correct match and a completely unknown animal produce practically indistinguishable scores. Correct confident decisions, by contrast, score 0.966 to 0.976 with a clear gap to the runner-up. The system decides reliably only when a similarly-posed gallery image pushes the score above roughly 0.95; otherwise everything concentrates in the same 0.84 to 0.88 band.

We caution that the 100% unregistered recall is obtained from only 49 unregistered queries with a threshold tuned on the same set, and must not be read as a guarantee of perfect rejection on new data.

VI. DISCUSSION

Three implications follow.

Tune the backbone learning rate first. For ViT-based animal Re-ID, the standard conservative fine-tuning rate of 10^{-5} substantially under-adapts a self-supervised backbone

to a fine-grained wildlife domain. Practitioners with a limited compute budget should sweep η_{bb} before touching loss hyperparameters, whose combined effect here is an order of magnitude smaller.

Reconsider the ArcFace-plus-triplet default. The combination is near-universal in Re-ID, but at a few hundred training identities the triplet term monotonically degraded performance. Its benefit likely requires either more identities or a backbone that has not already been pretrained toward strong local discrimination.

Open-set rejection is the deployment bottleneck. The representation transfers; the decision rule does not. Because failure is in score calibration rather than embedding quality, the productive directions are calibration and score normalization, for instance distance normalization against a cohort, learned rejection heads, or explicit training-time exposure to unenrolled individuals, rather than further backbone scaling. Rejection thresholds must additionally be tuned on a held-out set, which our protocol did not do.

The main methodological limitation is single-seed evaluation. Parameters with ranges below two points cannot be separated from run-to-run variance, and the exact match between two configurations across three metrics observed for $\gamma = 0.85$ warrants verification against experiment logs. OFAT also evaluates each parameter at an operating point that Table III shows to be under-trained, so some measured effects are conditioned on a non-converged reference.

VII. Conclusion

We presented a DINOv3 ViT sea turtle Re-ID system evaluated under open-set identity splits, together with a sensitivity map over nineteen parameters. A convention-based baseline reaches Rank-1 of 78.3% and mAP of 84.9%; combining individually-best settings reaches 90.4% and 93.5%, with k-reciprocal re-ranking contributing about one point in both cases. Backbone learning rate is the dominant factor, contributing roughly 78% of the total improvement on its own, while six parameters move Rank-1 by under two points. Parameter effects are strongly sub-additive: only 36% of the additively predicted gain materializes, and the additive prediction exceeds 100%, so OFAT should be used to order parameters by importance, never to forecast combined performance.

The embedding transfers to an unseen capture domain at Rank-1 of 81.1% and Rank-5 of 97.2% without retraining. Open-set rejection, however, reduces effective accuracy to 55–57% at a false-reject rate near 42%, and two structurally different rejection rules converge on nearly identical operating points because top-1 similarities occupy a narrow band with ROC-AUC of only 0.77–0.79. Since correct identities remain in the top five even in the worst failures, the limitation is calibration, not representation. Future work should pursue multi-seed evaluation with factorial verification, held-out threshold tuning with more unenrolled identities, exploitation of left-right profile similarity, integration of head detection into an end-to-end pipeline, and validation on species beyond loggerhead.

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